

Ilya Venediktov

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EDUCATION

Ph.D. in Physics

Ohio University, Athens, OH, USA

Expected: 05/2026

Dissertation: “Weak Gravitational Lensing and the LoVoCCS Survey: Empirical Bias Calibration with HST”

- Advisor: Dr. Douglas Clowe
- Focus: Weak Gravitational Lensing

GPA: 3.776/4

M.S. in Physics

Ohio University, Athens, OH, USA

Graduation 08/2022

GPA: 3.735/4

B.S in Astrophysics

Rutgers University, New Brunswick, NJ, USA

- Dean’s List (4 semesters)

Graduation 05/2020

GPA: 3.333/4

A.S in Physics

Brookdale Community College, Lincroft, NJ, USA

- Dean’s list (3 semesters), Phi Theta Kappa Honor Society, Distinguished Scholar Award

Graduation 06/2015

GPA: 3.81/4

RESEARCH EXPERIENCE

Research Assistant

Department of Physics and Astronomy, Ohio University

August, 2022 – Present

- Conduct experimental/theoretical research in [area of focus] under NSF/DOE grant support
- Developed and implemented simulation models using [e.g., Python, MATLAB, C++]
- Presented findings at regional and national conferences

Graduate Research Preparation

Department of Physics & Astronomy, Ohio University

2024–2025

- Prepared a dissertation prospectus titled Weak Gravitational Lensing and the LoVoCCS Survey: Empirical Bias Calibration with HST
- Reviewed literature and outlined methodology for calibrating multiplicative shear bias using matched HST and DECam data
- Explored use of SExtractor, PSFEx, GalSim, and LSST Science Pipelines for weak lensing shape measurement and simulation
- Developed preliminary workflow for catalog cross-matching and shear comparison

- Gained experience in proposal writing, literature synthesis, and planning of observational cosmology research

TEACHING EXPERIENCE

Teaching Assistant

Department of Physics and Astronomy, Ohio University

August, 2020– August, 2022

- Instructed lab sections for Introduction to Physics (PHYS 2001)
- Held weekly office hours and provided individualized support
- Graded labs/exams

Grader

Department of Physics and Astronomy, Ohio University

August, 2021– December, 2021

- Graded homework assignments for Introduction to Quantum Mechanics (PHYS 4021)

WORK EXPERIENCE

Assistant

Imperium Construction Inc, Brooklyn, NY, USA

May, 2018 – August, 2018

- Prepared cost estimate sheets for construction contracts by collecting information about construction materials

TECHNICAL SKILLS

- Programming: Python (NumPy, SciPy, Astropy, Matplotlib), C/C++, Bash, Perl, IDL, Jupyter Notebooks
- Data Analysis & Visualization: SExtractor, PSFEx, FITS file handling (Astropy, DS9), LaTeX
- Astronomical Tools: Catalog cross-matching, GalSim (basic), LSST Science Pipelines (introductory), run steps pipeline for LoVoCCS (introductory)

AWARDS & HONORS

- Phi Theta Kappa Honor Society , Brookdale Community College
- Distinguished Scholar Award, Brookdale Community College
- Dean's List, Brookdale Community College (Spring 2014, Fall 2014, Spring 2015)
- Dean's List, Rutgers University (Fall 2017, Spring 2018, Spring 2019, Spring 2020)

RELEVANT GRADUATE COURSEWORK

PHYS 5041 – Mathematical Methods in Physics I

- Covers complex variables, vector calculus, linear algebra, Fourier analysis, and partial differential equations.

- Learning outcomes: Apply advanced mathematical tools to solve theoretical physics problems.

PHYS 6001 – Classical Mechanics

- Introduces Newtonian mechanics, Lagrangian and Hamiltonian formalisms, central force motion, and rigid body dynamics.
- Learning outcomes: Develop problem-solving skills in classical systems using variational principles and conservation laws.

PHYS 6020 – Quantum Mechanics I

- Focuses on wave mechanics, Schrödinger equation, operators, eigenvalue problems, and potential wells.
- Learning outcomes: Understand and solve basic quantum systems; interpret quantum postulates.

PHYS 6021 – Quantum Mechanics II

- Continues from Quantum I includes angular momentum, spin, perturbation theory, and identical particles.
- Learning outcomes: Apply approximation techniques to multi-dimensional and realistic quantum systems.

PHYS 6011 – Statistical Mechanics I

- Introduces microcanonical, canonical, and grand canonical ensembles, partition functions and thermodynamic quantities.
- Learning outcomes: Derive macroscopic properties from microscopic models using statistical methods.

PHYS 6031 – Electrodynamics I

- Covers electrostatics, magnetostatics, Maxwell's equations, boundary-value problems, and wave propagation.
- Learning outcomes: Solve classical electromagnetic problems using vector calculus and boundary conditions.

PHYS 5071 – Computer Simulation Methods in Physics

- Introduces numerical methods including Monte Carlo techniques and differential equation solvers.
- Learning outcomes: Develop and apply simulations to model physical systems computationally.

PHYS 6940 – Special Study

- Individual research projects in physics, conducted under faculty supervision.
- Learning outcomes: Gain experience in problem formulation, literature review, and data/code-driven research.

PHYS 8201 – Research Seminar: Astrophysics

- Graduate-level seminar presenting current research topics in astrophysics.
- Learning outcomes: Practice scientific presentation and critical reading of research literature.

ASTR 5271 – Observational Astrophysics

- Covers astronomical instrumentation, CCD imaging, photometry, and data reduction techniques.
- Learning outcomes: Conduct and analyze observations with research-grade instruments.

ASTR 5201 – Stellar Astrophysics and Radiation

- Covers stellar interiors, nuclear processes, radiative transfer, and spectral analysis.
- Learning outcomes: Understand physical principles governing stellar structure and evolution.

ASTR 5202 – Interstellar Medium and Galaxies

- Explores phases of the interstellar medium, star formation, galactic structure, and dynamics.
- Learning outcomes: Analyze how gas dynamics and stellar processes shape galaxies.

PHYS 7021 – Quantum Field Theory I (In Progress)

- Introduces quantum field theory, covering canonical quantization, scalar and spinor fields, and Feynman diagrams.
- Learning outcomes: Understand the quantization of fields and apply perturbative techniques to quantum interactions.

RELEVANT UNDERGRADUATE COURSEWORK

Thermal Physics (01:750:351)

- Explores laws of thermodynamics and basics of statistical mechanics.
- Learning outcomes: Understand entropy, thermodynamic potentials, and distributions of microstates.

Quantum and Atomic Physics (01:750:361)

- Introduces quantum theory and atomic structure: wavefunctions, quantization, and hydrogen atom.
- Learning outcomes: Solve quantum problems and interpret physical results at atomic scales.

Mechanics II (01:750:382)

- Advanced study of Lagrangian and Hamiltonian mechanics, oscillations, and central force problems.
- Learning outcomes: Analyze motion in non-inertial frames and develop fluency with variational principles.

Electromagnetism I (01:750:385)

- Intermediate-level course covering electrostatics, electric potential, Laplace's equation, method of images, multipole expansion, electric fields in matter, polarization, dielectrics, magnetostatics, Biot–Savart law, Ampère's law, magnetic vector potential, and magnetized materials.
- Learning outcomes: Develop proficiency in solving boundary-value problems in electrostatics and magnetostatics; understand the behavior of electric and magnetic fields in various media.

Electromagnetism II (01:750:386)

- Continuation of Electromagnetism I, focusing on electrodynamics, including electromagnetic induction, Maxwell's equations, electromagnetic waves, energy and momentum conservation, scalar and vector potentials, radiation, and the relationship between electrodynamics and special relativity.
- Learning outcomes: Apply Maxwell's equations to dynamic systems; analyze electromagnetic wave propagation; understand the relativistic formulation of electromagnetism.

Principles of Modern Physics (01:750:341)

- Surveys early 20th-century physics: relativity, quantum theory, nuclear and particle physics.
- Learning outcomes: Understand the foundations of modern physical theories.

Principles of Astrophysics (01:750:342)

- Covers stellar structure, radiative transfer, celestial mechanics, and the interstellar medium.

- Learning outcomes: Apply physics principles to astrophysical systems.

Observational Radio Astronomy (01:750:343)

- Introduces techniques for acquiring and analyzing radio frequency astronomical data.
- Learning outcomes: Operate radio telescopes and reduce observational datasets.

Observational Optical Astronomy (01:750:344)

- Covers photometry, CCD imaging, and optical data analysis.
- Learning outcomes: Conduct optical observations and extract scientific measurements.

Stars and Star Formation (01:750:441)

- Covers stellar physics: nuclear reactions, structure, evolution, and star-forming regions.
- Learning outcomes: Explain stellar lifecycle processes and compare theory to observations.

Galaxies and the Milky Way (01:750:443)

- Studies the structure, dynamics, and formation of galaxies.
- Learning outcomes: Interpret galactic observational data and understand large-scale evolution.

Introduction to Cosmology (01:750:444)

- Covers expanding universe, cosmic microwave background, and large-scale structure.
- Learning outcomes: Describe the standard cosmological model and interpret cosmological observations.

Computational Astrophysics (01:750:345)

- Project-based study in computational techniques for astrophysics.
- Learning outcomes: Apply code and simulations to astrophysical systems.

References

Dr. Refer Enceone	Dr. Secon Dereference
Professor	Professor
Psychology Department	Criminology
45 Porter Hall	56 Porter Hall
Sample University	Sample University
Lovely, OH	Lovely, OH
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